

SECTION A**1 (long)**

- (a) State Thevenin's and Norton's theorems. [3]
- (b) Calculate the Thevenin and Norton equivalents of the circuit in Fig. 1(a). [9]
- (c) Figure 1(b) shows the equivalent circuit of a solar cell under illumination by sunlight. It consists of a 50 mA ideal current source in parallel with a diode and a $12\ \Omega$ parasitic shunt resistance. If the current-voltage characteristics of the diode are as shown in Fig. 2, calculate the output voltage, V_o , between the cell's terminals A' and B'. *A copy of Fig. 2. is reproduced at the end of this paper for you to detach and hand in with your answer.* [9]
- (d) The solar cell in Fig. 1(b) is connected to the circuit in Fig. 1(a) by connecting terminal A to terminal A' and terminal B to terminal B'. Calculate the new output voltage V_o from the solar cell. [9]

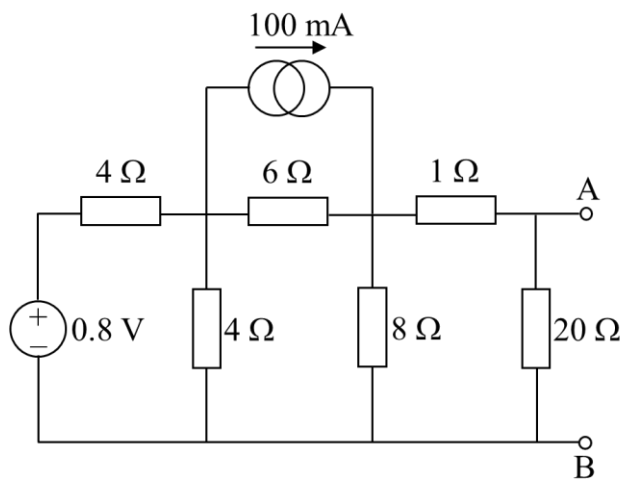


Fig. 1(a)

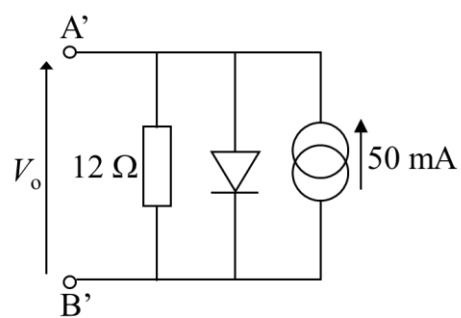


Fig. 1(b)

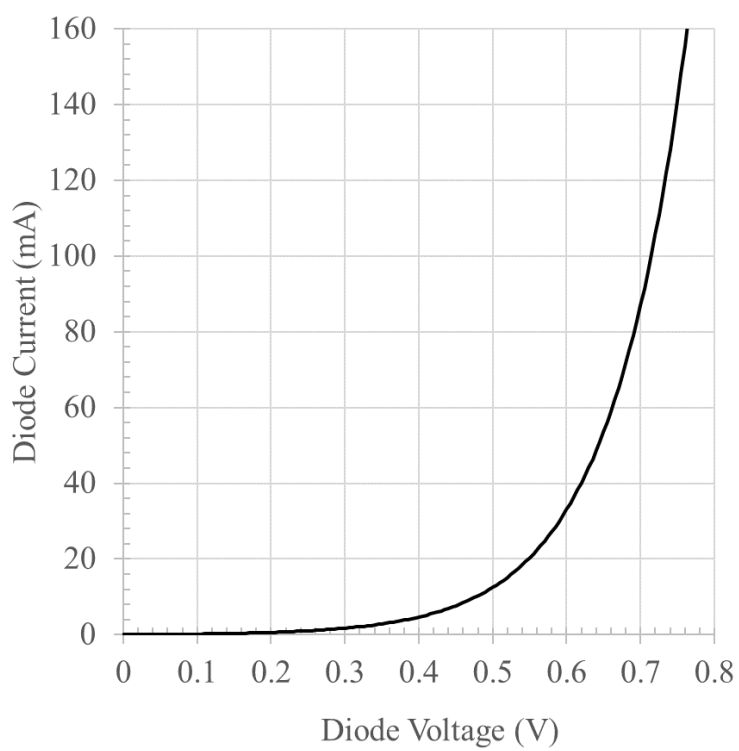


Fig. 2